

# Data Preprocessing & Cleaning

Data Mining (IF211)

Preprocessing & EDA

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## Preprocessing

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**"Garbage in, garbage out."** Preprocessing mengubah data mentah menjadi format yang bisa diproses algoritma.

## Langkah

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1. **Missing Values:** Drop, fill mean/median/mode, interpolate
2. **Encoding:** Label encoding (ordinal), one-hot (nominal)
3. **Scaling:** Standardization, min-max, robust
4. **Outliers:** IQR method, z-score

```
df['age'].fillna(df['age'].median(), inplace=True)
scaler = StandardScaler()
df[['income', 'score']] = scaler.fit_transform(df[['income', 'score']])
```

## Ringkasan

Handle missing → encode → scale → detect outliers. Split train/test SEBELUM scaling. Pilihan metode tergantung distribusi dan algoritma.